Grade 10–12 Mathematics

DBE CAPS Comprehensive Handbook & Textbook Outline

Grade 10

Paper 1

Topic / Module	Core CAPS Curriculum Requirements
Algebra & Equations	Exponents & laws; Factorisation (trinomials, cubes); Solving linear, quadratic & simultaneous equations; Linear inequalities.
Number Patterns	Investigating, extending, and finding the general formula (n-th term) for linear/arithmetic sequences.
Functions & Graphs	Sketching & interpreting: Linear (y=mx+c), Quadratic ($y=ax^2$), Hyperbolic (y=a/x), and Exponential (y=ab^x).
Finance & Growth	Simple and compound interest; Simple & compound depreciation; Hire-purchase; Exchange rates.
Probability	Venn diagrams; Theoretical & relative frequency probability; Identity rule / addition rule.

Paper 2

Topic / Module	Core CAPS Curriculum Requirements
Trigonometry	Trig ratios in right-angled triangles; Cartesian plane (0° to 360°); Two-dimensional problems; Basic graphs (sin, cos, tan).
Euclidean Geometry	Lines, angles, triangles; Properties of quadrilaterals (parallelograms, rectangles, rhombi, kites); Midpoint Theorem.
Analytical Geometry	Distance formula; Midpoint formula; Gradient/slope of a line segment on the Cartesian plane.
Statistics	Data collection & organization; Central tendency (mean, median, mode); Dispersion (range, quartiles, percentiles); Box plots.

Grade 11

Paper 1

Topic / Module	Core CAPS Curriculum Requirements
Algebra & Equations	Surds and fractional exponents; Completing the square; Quadratic formula; Nature of roots; Quadratic inequalities; Simultaneous equations (linear + quadratic).
Number Patterns	Quadratic number patterns: finding the second constant difference and the general term ($T_n = an^2 + bn + c$).
Functions & Graphs	Advanced transformations & sketching: $y = a(x-p)^2 + q$, $y = a/(x-p) + q$, and $y = ab^{(x-p)} + q$. Inverse functions introduction.
Finance & Growth	Sinking funds; Present value and Future value annuities introduction; Nominal vs. Effective interest rates.
Probability	Dependent and independent events; Product rule for independent events; Tree diagrams; Contingency tables.

Paper 2

Topic / Module	Core CAPS Curriculum Requirements
Trigonometry	Reduction formulae to the 4 quadrants; Co-functions; Trigonometric identities ($\tan = \sin/\cos$, $\sin^2+\cos^2=1$); General solutions; Sine, Cosine, and Area rules.
Euclidean Geometry	Circle Geometry theorems: chords, angles at center/circumference, cyclic quadrilaterals, tangents to circles.
Analytical Geometry	Inclination of a line; Parallel and perpendicular line properties; Solving complex geometric shapes analytically.
Statistics	Grouped data analysis; Ogives (cumulative frequency graphs); Variance and standard deviation; Symmetric and skewed data.

Grade 12

Paper 1

Topic / Module	Core CAPS Curriculum Requirements
Algebra & Equations	Review of quadratic equations & inequalities; Advanced simultaneous equations; Logarithms application in exponential equations.
Patterns & Series	Arithmetic and Geometric sequences & series; Sigma notation; Sum to infinity for convergent geometric series.
Functions & Graphs	Concepts of domain & range; Formal definition of inverse functions; Restricting domain for a quadratic function's inverse ($y = \sqrt{x}$).
Calculus	Limits definition; Differentiation from first principles; Rules of differentiation; Cubic graphs; Optimization and rates of change.
Finance & Growth	Full application of Future Value and Present Value annuity formulae; Calculating loan repayments and outstanding balances.
Probability	Fundamental Counting Principle; Permutations; Factorial notation; Application of counting principles to probability.

Paper 2

Topic / Module	Core CAPS Curriculum Requirements
Trigonometry	Compound angle and Double angle identities; Expansion and simplification; Application of rules in 3D geometric settings.
Euclidean Geometry	Proportionality in triangles; Similarity of triangles; Theorem of Pythagoras proven via similarity.
Analytical Geometry	Equation of a circle (centered at origin or any point); Tangents to circles analytically; Intersecting circles.
Statistics	Bivariate data; Scatter plots and lines of best fit; Least-squares regression line ($y = A + Bx$); Correlation coefficient (r).